Type MS · Trace the route of blood through the heart, lungs and body

Answer 1 of 9 · Question 5 part a · 5 marks

2023 JUN · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

5 (a)	*hepatic vein* vena cava; pulmonary artery; pulmonary vein; aorta; renal artery;	Credit correct order	5
(b) (i)	Any four from: • as time spent exercising increases, pulse rate increases; • increase flow of blood (to muscles/cells); • more oxygen/glucose/energy; • more (aerobic) respiration; • more carbon dioxide removed/reference to reduced build-up of lactic acid;	Allow more blood pumped Allow less anaerobic respiration	4
(ii)	Any six from • take pulse before exercise/resting pulse; • finger at wrist/neck; • carry out exercise; • count/record pulse (after exercise); • wait for pulse rate to drop back to normal; • repeat exercise; • reference to control variables e.g. same person/gender/exercise/intensity of exercise/weather conditions;	Allow reference to use of electronic device to measure pulse rate Allow adequate time frame e.g. 5 minutes Allow reference to reliability e.g. more people	6

Total 15

Type MS · Trace the route of blood through the heart, lungs and body

Answer 1 of 9 · Question 5 part a · 5 marks

2023 JUN · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Trace the route of blood through the heart, lungs and body

Answer 2 of 9 · Question 1 part a.iii · 2 marks

2022 JAN · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

Question number					
1	(a)	(i)	B; (pulmonary artery) *A it is not the aorta* *C it is not the pulmonary vein* *D it is not the vena cava*		1
		(ii)	A; (aorta) *B it is not the pulmonary artery* *C it is not the pulmonary vein* *D it is not the vena cava*		1
		(iii)	in at top left and out through vessel; in at top right and out through vessel;		2
		(iv)	valves; prevent blood flowing other way/backflow;		2
	(b)	(i)	chamber width in cm P 1.0; Q 0.2; R 0.2; S 1.5;	1 correct = 1 mark 2 correct = 2 marks 3/4 correct = 3 marks	3
		(ii)	An answer that makes reference to three of the following: • The width corresponds to the amount of muscle; • S pumps blood furthest distance/whole body; • P pumps blood to lungs/short distance; • Q/R only receives blood/doesn't have to pump it very far/to ventricles;		3

Total : 12 marks

Type MS · Trace the route of blood through the heart, lungs and body

Answer 2 of 9 · Question 1 part a.iii · 2 marks

2022 JAN · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Trace the route of blood through the heart, lungs and body

Answer 3 of 9 · Question 1 part b · 5 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

1 (a)	pulmonary artery;		1
(b)	In the following order: • oxygenated; • lungs; • atrium; • ventricle; • body;		5
(c)	• blood vessel X has a thicker wall; • blood vessel X has a smaller lumen; • blood vessel X does not contain valves	Allow the reverse argument for blood vessel Y	3
			Total 9

Type MS · Trace the route of blood through the heart, lungs and body

Answer 3 of 9 · Question 1 part b · 5 marks

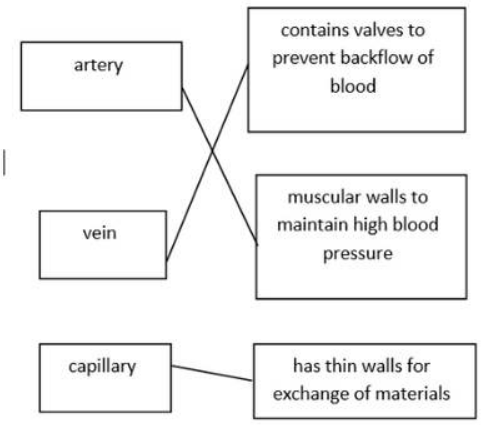
2022 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Trace the route of blood through the heart, lungs and body

Answer 4 of 9 · Question 2 part c.ii · 4 marks

2024 JUN · Paper 01 (regional R) · find part c.ii in this mark scheme extract

2 (a) (i)	X = plasma; Y = red blood cell; Z = white blood cell/phagocyte;	Accept named phagocyte	(3)
(a) (ii)	• transports oxygen; • to body/ cells/tissues/organs;		(2)
(a) (iii)	the patient has an infection/leukaemia/allergy/ blood cancer, inflammation;	Accept pathogens in body	(1)
(b)		Reject more than one line from each blood vessel 3 correct=2marks 1 or 2 correct= 1mark	(2)
(c) (i)	left ventricle;		(1)
(c) (ii)	• into right ventricle; • through pulmonary artery; • to the lungs; • (from the lungs) into the pulmonary vein;		(4)

Total for Question 2 = 13 marks

Type MS · Trace the route of blood through the heart, lungs and body

Answer 5 of 9 · Question 5 part b · 3 marks

2024 JUN · Paper 02 (regional R) · find part b in this mark scheme extract

5 (a)	(cardiac) muscle(s);		(1)
(b)	• pumps blood twice in one circuit; • right side pumps deoxygenated blood to lungs; • left side pumps oxygenated blood to body;		(3)
(c)	**C (lymphocytes and phagocytes)** The following are incorrect because: A platelets are not cells B antibodies and hormones are not cells D platelets are not cells		(1)
(d) (i)	hormones;		(1)
(ii)	platelets/fibrinogen/prothrombin;		(1)
(e)	(amount of blood in 1kg) 5.5 ÷ 70 = 0.07857; (amount of blood in 95kg) 0.07857 x 95 = 7.464; (number of red blood cells in 95kg) = 7.464 x 5.9 x 10^{12} = 4.4 x 10^{13};	allow 0.08 allow 7.5 allow 4.4038 ×10^{13} allow 4.425 × 10^{13} allow 4.484 × 10^{13} full marks for correct final answer	(3)

Total for Question 5 = 10 marks

Type MS · Trace the route of blood through the heart, lungs and body

Answer 6 of 9 · Question 1 part a · 8 marks

2025 JUN · Paper 01 (regional R) · find part a in this mark scheme extract

1 (a)	• atria; • lungs; • vena cava; • ventricles; • body; • muscle; • systolic; • 120;		8
(b)	6.1 ÷ 68; = 0.0897 dm^3; = 89.7 (cm^3);	ACCEPT 90 (cm^3) Allow 1 mark for incorrect dm^3 in MP2 to correct conversion to cm^3	3
			Total = 11

Question number	Answer	Notes	Marks

Type MS · Trace the route of blood through the heart, lungs and body

Answer 7 of 9 · Question 4 part b · 4 marks

2020 JAN · Paper 01 · find part b in this mark scheme extract

Question number				
4	(a)	• C (left) atrium(1) • D (left) ventricle (1)	Maximum of one mark if right atrium/ventricle given	2
	(b)	Any four from the following: • (blood vessels) A and F are arteries (1) • A is the aorta/F is the pulmonary artery (1) • arteries carry blood away from the heart (1) • blood vessels B and E are veins (1) • B is the vena cava/E is the pulmonary vein (1) • veins carry blood towards the heart (1)		4
	(c)	the heart pumps blood around the body/liver detoxifies the blood/involved in digestion / homeostasis/liver not muscular/liver has no chambers (1)		1
	(d)	(capillaries) exchange of materials(between the blood and body tissues/cells)/allows diffusion (1) named substance (1)		2

Total for Question 4 = 9 marks

Type MS · Trace the route of blood through the heart, lungs and body

Answer 8 of 9 · Question 3 part a.iii · 3 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part a.iii in this mark scheme extract

3 (a)	(i)	250:750; 1:3;	full marks for correct answer with no working	2
	(ii)	17 500 - 5600 = 11900; $\frac{11\,900 \times 100}{5\,600}$; = 210%;	full marks for correct answer with no working 213/212.5 = two marks	3
	(iii)	• other parts of the body/organs; • liver/kidney/lungs/arteries/veins; • needs some blood supplied/blood flows to these organs;		3
(b)	(i)	• large/big/huge increase in total flow; • large/big/huge increase in flow skeletal muscles/skin; • decrease for gut;		3
	(ii)	• heat generated(body) temperature increases during exercise; • from respiration; • carried by blood/more blood to skin; • (heat) lost through skin/by radiation;		4

Total 15 marks

Type MS · Trace the route of blood through the heart, lungs and body

Answer 8 of 9 · Question 3 part a.iii · 3 marks

2023 NOV · Paper 01 · extract 2 of 2 · find part a.iii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Trace the route of blood through the heart, lungs and body

Answer 9 of 9 · Question 4 part c · 1 mark

2020 JAN · Paper 01 · find part c in this mark scheme extract

Number				
4	(a)	• C (left) atrium(1) • D (left) ventricle (1)	Maximum of one mark if right atrium/ventricle given	2
	(b)	Any four from the following: • (blood vessels) A and F are arteries (1) • A is the aorta/F is the pulmonary artery (1) • arteries carry blood away from the heart (1) • blood vessels B and E are veins (1) • B is the vena cava/E is the pulmonary vein (1) • veins carry blood towards the heart (1)		4
	(c)	the heart pumps blood around the body/liver detoxifies the blood/involved in digestion / homeostasis/liver not muscular/liver has no chambers (1)		1
	(d)	(capillaries) exchange of materials(between the blood and body tissues/cells)/allows diffusion (1) named substance (1)		2

Total for Question 4 = 9 marks